Chapter 2, Problem 8RQ

Problem

In the circuit in Fig. 2.65, V is:

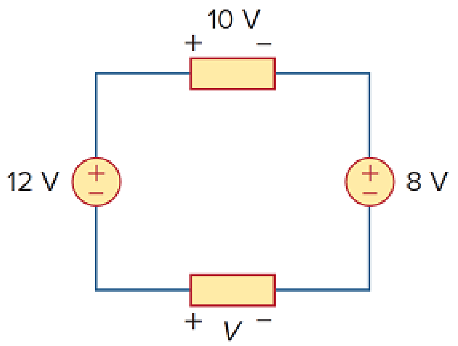
(a) 30 V

(b) 14 V

(c) 10 V

(d) 6 V

Figure 2.65:

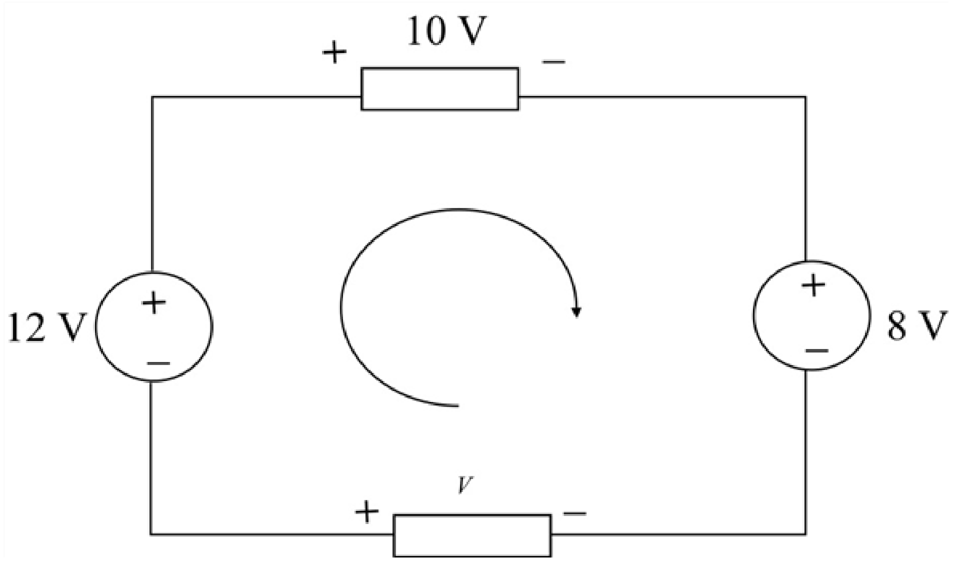


Step-by-step solution

Step 1 of 2

Refer to the circuit diagram in Figure 2.65 in the textbook.

Redraw the circuit diagram.



Step 2 of 2

Apply Kirchhoff's voltage law around loop.

$$-12\text{ V}+10\text{ V}+8\text{ V}-V=0$$
$$V=6\text{ V}$$

Thus, correct option is $\boxed{d}$ or **6 V.**